

Lecture 3: Concentration Inequalities

MGF Method, Hoeffding, Bernstein,
and the Sub-Gaussian Family

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1 Recap and Roadmap

In Lectures 1–2 we saw that the asymptotic paradigm provides sharp, interpretable guarantees under regularity conditions, but breaks down under misspecification, heavy tails, or high dimensions. The non-asymptotic paradigm offers an alternative: *explicit probability bounds valid for every fixed n* , with minimal distributional assumptions.

Lecture 1 introduced Chebyshev’s inequality, whose tail bound decays as $O(1/n)$. Today we develop the **MGF method**, which produces *exponentially* decaying bounds — a dramatic improvement.

Lecture 3 roadmap:

1. **Cramér–Chernoff method**: one-sided exponential bounds via MGF optimization.
2. **Hoeffding’s inequality**: the canonical bound for bounded random variables; full proof.
3. **Bernstein’s inequality and the sub-Gaussian/sub-exponential families**: variance-aware bounds and a unifying framework.
4. **Tightness, limitations, and comparison with CLT**.

The Central Strategy of This Lecture

Every bound derived today follows the same three-step recipe:

1. Apply Markov’s inequality to e^{tX} for $t > 0$.
2. Bound the MGF $\mathbb{E}[e^{tX}]$ using the structure of X .
3. Optimize over t to get the sharpest bound.

Understanding this template is more valuable than memorizing any single inequality.

2 The Cramér–Chernoff Method

2.1 One-sided bounds via the MGF

Let Z be a real-valued random variable with MGF $M_Z(t) = \mathbb{E}[e^{tZ}]$ finite for t in a neighborhood of 0.

Theorem 2.1 — Cramér–Chernoff Bound

For any $\varepsilon > 0$ and $t > 0$,

$$\mathbb{P}(Z \geq \varepsilon) \leq e^{-t\varepsilon} M_Z(t).$$

Optimizing over $t > 0$:

$$\mathbb{P}(Z \geq \varepsilon) \leq \inf_{t>0} e^{-t\varepsilon} M_Z(t) =: e^{-\Lambda^*(\varepsilon)},$$

where $\Lambda^*(\varepsilon) = \sup_{t>0} [t\varepsilon - \log M_Z(t)]$ is the **Legendre–Fenchel transform** (convex conjugate) of the cumulant generating function $\Lambda(t) = \log M_Z(t)$.

Proof. For any $t > 0$, $\{Z \geq \varepsilon\} = \{e^{tZ} \geq e^{t\varepsilon}\}$, so by Markov’s inequality:

$$\mathbb{P}(Z \geq \varepsilon) = \mathbb{P}(e^{tZ} \geq e^{t\varepsilon}) \leq \frac{\mathbb{E}[e^{tZ}]}{e^{t\varepsilon}} = e^{-t\varepsilon} M_Z(t).$$

Since this holds for every $t > 0$, we may take the infimum over t . □

Remark — Why Exponential Tails?

Chebyshev uses the second moment: $\mathbb{P}(Z \geq \varepsilon) \leq \text{Var}(Z)/\varepsilon^2$, which decays as $O(\varepsilon^{-2})$. The Cramér–Chernoff bound uses *all* moments simultaneously via the MGF, yielding a bound that decays *exponentially* in ε . This is the fundamental gain of the MGF method.

2.2 Application to sums of independent variables

The MGF method is especially powerful for sums, because independence factorizes the MGF:

Lemma 2.1 (MGF of a sum). *If $X_1, \dots, X_n$ are independent, then*

$$M_{X_1 + \dots + X_n}(t) = \prod_{i=1}^n M_{X_i}(t).$$

Proof. By independence, $\mathbb{E}[\exp(t \sum_i X_i)] = \mathbb{E}[\prod_i e^{tX_i}] = \prod_i \mathbb{E}[e^{tX_i}]$. □

Example 2.3 — Gaussian Sum

Let $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. The MGF is $M_{X_i}(t) = e^{\sigma^2 t^2/2}$, so $M_{\bar{X}_n}(t) = e^{\sigma^2 t^2/(2n)}$. The Cramér–Chernoff bound gives, for $\varepsilon > 0$:

$$\mathbb{P}(\bar{X}_n \geq \varepsilon) \leq \inf_{t>0} \exp\left(-t\varepsilon + \frac{\sigma^2 t^2}{2n}\right).$$

Minimizing over t : optimal $t^* = n\varepsilon/\sigma^2$, yielding

$$\mathbb{P}(\bar{X}_n \geq \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

This is *exact* for Gaussians (matches the Gaussian tail probability up to constants), confirming the bound is tight in this case.

3 Hoeffding’s Inequality**3.1 Hoeffding’s lemma: bounding the MGF of a bounded variable**

The key ingredient is a tight upper bound on $\log M_X(t)$ when X is bounded.

Lemma 3.1 — Hoeffding's Lemma

Let X be a random variable with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely. Then for all $t \in \mathbb{R}$,

$$\log \mathbb{E}[e^{tX}] \leq \frac{t^2(b-a)^2}{8}.$$

Proof. Since e^{tx} is convex in x , for $x \in [a, b]$ we have the linear upper bound:

$$e^{tx} \leq \frac{b-x}{b-a} e^{ta} + \frac{x-a}{b-a} e^{tb}.$$

Taking expectations (using $\mathbb{E}[X] = 0$):

$$\mathbb{E}[e^{tX}] \leq \frac{b}{b-a} e^{ta} - \frac{a}{b-a} e^{tb} =: e^{\varphi(u)},$$

where $u = t(b-a)$ and $\varphi(u) = -\gamma u + \log(1-\gamma+\gamma e^u)$ with $\gamma = -a/(b-a) \in [0, 1]$. A Taylor expansion of φ around $u = 0$ shows $\varphi(0) = \varphi'(0) = 0$ and $\varphi''(u) \leq 1/4$ for all u (since $\varphi''(u) = \gamma(1-\gamma)e^u/(1-\gamma+\gamma e^u)^2 \leq 1/4$ by AM-GM). Thus $\varphi(u) \leq u^2/8 = t^2(b-a)^2/8$. $\square$

Remark — The Constant 1/8

The constant 1/8 is tight: for $X = \pm(b-a)/2$ with equal probability and $a = -(b-a)/2$, $b = (b-a)/2$, one can verify $\log \mathbb{E}[e^{tX}] = \log \cosh(t(b-a)/2) \approx t^2(b-a)^2/8$ for small t .

3.2 Hoeffding's inequality**Theorem 3.2 — Hoeffding's Inequality**

Let $X_1, \dots, X_n$ be *independent* (not necessarily identically distributed) random variables with $\mathbb{E}[X_i] = \mu_i$ and $a_i \leq X_i \leq b_i$ a.s. Let $S_n = \sum_{i=1}^n X_i$ and $\mu = \sum_{i=1}^n \mu_i$. Then for all $\varepsilon > 0$,

$$\mathbb{P}(S_n - \mu \geq \varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Consequently, the two-sided version:

$$\mathbb{P}(|S_n - \mu| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Proof. Let $Y_i = X_i - \mu_i$, so $\mathbb{E}[Y_i] = 0$ and $Y_i \in [a_i - \mu_i, b_i - \mu_i]$. By the Cramér–Chernoff method and Lemma 2.1:

$$\mathbb{P}(S_n - \mu \geq \varepsilon) \leq e^{-t\varepsilon} \prod_{i=1}^n M_{Y_i}(t).$$

By Hoeffding's Lemma (Lemma 3.1), $\log M_{Y_i}(t) \leq t^2(b_i - a_i)^2/8$, so

$$\mathbb{P}(S_n - \mu \geq \varepsilon) \leq \exp\left(-t\varepsilon + \frac{t^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right).$$

Minimizing over $t > 0$: the exponent $-t\varepsilon + ct^2$ (with $c = \frac{1}{8} \sum (b_i - a_i)^2$) is minimized at $t^* = \varepsilon/(2c)$, giving

$$\mathbb{P}(S_n - \mu \geq \varepsilon) \leq \exp\left(-\frac{\varepsilon^2}{4c}\right) = \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

The two-sided bound follows by applying the one-sided bound to $S_n - \mu$ and $-(S_n - \mu)$ and taking a union bound. $\square$

Corollary 3.1 (Hoeffding for i.i.d. bounded variables). *Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ with $\mathbb{E}[X_i] = \mu$ and $a \leq X_i \leq b$ a.s. Then*

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Proof. Apply Theorem 3.2 with $S_n = n\bar{X}_n$, $a_i = a$, $b_i = b$ for all i , and replace ε by $n\varepsilon$. $\square$

Example 3.3 — Sample Proportion ($X_i \in [0, 1]$)

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$. By Corollary 3.1 with $a = 0$, $b = 1$:

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

Setting the right-hand side equal to δ and solving for ε : with probability at least $1 - \delta$,

$$|\hat{p}_n - p| \leq \sqrt{\frac{\log(2/\delta)}{2n}}.$$

For $\delta = 0.05$, $n = 1000$: the bound gives $|\hat{p}_n - p| \leq \sqrt{\log(40)/2000} \approx 0.042$. The CLT-based bound is $1.96\sqrt{p(1-p)/n} \leq 1.96/(2\sqrt{n}) \approx 0.031$.

Both bounds are $O(1/\sqrt{n})$, but Hoeffding is valid for *every* n with no approximation error; the CLT bound holds only approximately for large n .

4 Bernstein's Inequality and the Sub-Gaussian Family

4.1 Bernstein's inequality: exploiting the variance

Hoeffding's bound depends only on the range $[a, b]$ of each variable, ignoring the variance. When the variance is small relative to the range, Bernstein's inequality gives a tighter result.

Theorem 4.1 — Bernstein's Inequality

Let $X_1, \dots, X_n$ be independent with $\mathbb{E}[X_i] = 0$, $\text{Var}(X_i) = \sigma_i^2$, and $|X_i| \leq M$ a.s. for all i . Let $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. Then for all $\varepsilon > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \varepsilon\right) \leq \exp\left(-\frac{\varepsilon^2/2}{\sigma^2 + M\varepsilon/3}\right).$$

Proof. We use the MGF bound: for $|X| \leq M$ and $\mathbb{E}[X] = 0$, a Taylor expansion gives

$$\mathbb{E}[e^{tX}] \leq 1 + \frac{t^2 \sigma_i^2}{2(1 - tM)} \leq \exp\left(\frac{t^2 \sigma_i^2/2}{1 - tM}\right) \quad \text{for } 0 < t < 1/M.$$

(The first inequality uses $e^u \leq 1 + u + u^2/(2(1 - u))$ for $u = tX$ with $|tX| \leq tM < 1$; the second uses $1 + x \leq e^x$.) Summing over i and applying the Cramér–Chernoff bound:

$$\mathbb{P}\left(\sum_i X_i \geq \varepsilon\right) \leq \exp\left(-t\varepsilon + \frac{t^2 \sigma^2/2}{1 - tM}\right).$$

Setting $t = \varepsilon/(\sigma^2 + M\varepsilon) < 1/M$ and simplifying yields the stated bound. $\square$

Remark — Two Regimes of Bernstein's Inequality

The Bernstein bound has two regimes, controlled by the ratio $\varepsilon/(\sigma^2/M)$:

- **Gaussian regime** ($\varepsilon \ll \sigma^2/M$): the bound is approximately $\exp(-\varepsilon^2/(2\sigma^2/n))$, which matches the Gaussian tail — same as the CLT would give.
- **Poisson regime** ($\varepsilon \gg \sigma^2/M$): the bound is approximately $\exp(-\varepsilon/(2M))$, an exponential tail reflecting the boundedness of X_i .

Hoeffding's bound corresponds to treating all variables as if they were at the extreme values, effectively ignoring the variance. Bernstein improves this when $\sigma^2 \ll (b - a)^2$.

Example 4.2 — Bernstein vs. Hoeffding for Rare Events

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $p = 0.01$ (rare event), $n = 10,000$, $\varepsilon = 0.005$ (half the mean). Here $\sigma^2 = np(1-p) \approx 99$ and $M = 1$.

Hoeffding (Corollary 3.1): $2 \exp(-2 \cdot 10000 \cdot 0.005^2) = 2e^{-0.5} \approx 1.21$. *The Hoeffding bound exceeds 1 — it is vacuous!*

Bernstein (with $S_n = \sum X_i$, $\varepsilon' = n\varepsilon = 50$, $\sigma^2 = 99$, $M = 1$): $\exp(-50^2/(2 \cdot 99 + 50/3)) \approx \exp(-11.2) \approx 10^{-5}$.

When the variance is much smaller than the range, Bernstein is dramatically sharper.

4.2 The sub-Gaussian family

The Hoeffding and Gaussian bounds share the same functional form $\exp(-t^2\sigma^2/2)$ for the MGF. This motivates a unifying definition.

Definition 4.3 — Sub-Gaussian Random Variable

A centered random variable X (i.e., $\mathbb{E}[X] = 0$) is **sub-Gaussian** with parameter $\sigma > 0$, written $X \sim \text{subG}(\sigma^2)$, if

$$\mathbb{E}[e^{tX}] \leq e^{\sigma^2 t^2/2} \quad \text{for all } t \in \mathbb{R}.$$

The **sub-Gaussian norm** (Orlicz norm) is

$$\|X\|_{\psi_2} = \inf \left\{ K > 0 : \mathbb{E}[e^{X^2/K^2}] \leq 2 \right\}.$$

Proposition 4.4 — Equivalent Characterizations of Sub-Gaussian

For a centered random variable X , the following are equivalent (with constants changing by absolute factors):

- (i) MGF condition: $\mathbb{E}[e^{tX}] \leq e^{\sigma^2 t^2/2}$ for all $t \in \mathbb{R}$.
- (ii) Tail condition: $\mathbb{P}(|X| \geq u) \leq 2e^{-u^2/(2\sigma^2)}$ for all $u \geq 0$.
- (iii) Moment condition: $(\mathbb{E}[|X|^k])^{1/k} \leq C\sigma\sqrt{k}$ for all $k \geq 1$.
- (iv) Orlicz condition: $\mathbb{E}[e^{X^2/\sigma^2}] \leq 2$.

Proof sketch: (i) $\Rightarrow$ (ii). Apply the Cramér–Chernoff bound with parameter σ^2 : $\mathbb{P}(X \geq u) \leq e^{-tu}e^{\sigma^2 t^2/2}$. Optimizing at $t = u/\sigma^2$ gives $\mathbb{P}(X \geq u) \leq e^{-u^2/(2\sigma^2)}$. Applying the same to $-X$ (which is also sub-Gaussian) and taking a union bound yields (ii). The other directions use Markov’s inequality with e^{X^2/K^2} and moment generating properties of the Gaussian. $\square$

Example 4.5 — Examples of Sub-Gaussian Variables

- (a) **Gaussian:** $X \sim \mathcal{N}(0, \sigma^2)$ is $\text{subG}(\sigma^2)$. (The MGF condition holds with equality.)
- (b) **Bounded:** $X \in [a, b]$ with $\mathbb{E}[X] = 0$ is $\text{subG}\left(\frac{(b-a)^2}{4}\right)$ by Hoeffding's Lemma.
- (c) **Rademacher:** $X = \pm 1$ with equal probability is $\text{subG}(1)$.
- (d) **Not sub-Gaussian:** $X \sim \text{Exponential}(1) - 1$ (mean zero, unbounded above). The MGF $\mathbb{E}[e^{tX}] = e^{-t}/(1-t)$ grows faster than e^{ct^2} for any c , so $X \notin \text{subG}(\sigma^2)$ for any σ^2 .

Theorem 4.6 — Sub-Gaussian Tail Bound for Sums

Let $X_1, \dots, X_n$ be independent with $X_i \sim \text{subG}(\sigma_i^2)$. Then $\sum_{i=1}^n X_i \sim \text{subG}(\sum_{i=1}^n \sigma_i^2)$, and

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \varepsilon\right) \leq \exp\left(-\frac{\varepsilon^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

Proof. By independence (Lemma 2.1), $M_{\sum X_i}(t) = \prod_i M_{X_i}(t) \leq \prod_i e^{\sigma_i^2 t^2 / 2} = e^{(\sum_i \sigma_i^2) t^2 / 2}$. This is the sub-Gaussian MGF condition with parameter $\sum_i \sigma_i^2$. The tail bound then follows from the Cramér–Chernoff method. $\square$

4.3 The sub-exponential family

For variables with heavier tails (e.g., χ^2 , Poisson), the sub-Gaussian assumption is too strong.

Definition 4.7 — Sub-Exponential Random Variable

A centered random variable X is **sub-exponential** with parameters (ν^2, α) , written $X \sim \text{subE}(\nu^2, \alpha)$, if

$$\mathbb{E}[e^{tX}] \leq e^{\nu^2 t^2 / 2} \quad \text{for all } |t| \leq 1/\alpha.$$

Equivalently, X is sub-exponential if and only if X^2 is sub-exponential in the sense that $\mathbb{P}(|X| \geq u) \leq 2e^{-u/\alpha}$ for large u .

Theorem 4.8 — Bernstein-Type Bound for Sub-Exponential Variables

Let $X_1, \dots, X_n$ be independent sub-exponential random variables with parameters (ν_i^2, α) . Then for $0 \leq \varepsilon \leq \sum_i \nu_i^2 / (\alpha n)$:

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i \geq \varepsilon\right) \leq \exp\left(-\frac{n\varepsilon^2}{2 \sum_i \nu_i^2 / n}\right),$$

and for $\varepsilon > \sum_i \nu_i^2 / (\alpha n)$:

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i \geq \varepsilon\right) \leq \exp\left(-\frac{n\varepsilon}{2\alpha}\right).$$

Proof sketch. Write $S_n = \sum_{i=1}^n X_i$ and $V = \sum_{i=1}^n \nu_i^2$, so the event of interest is $\{S_n \geq n\varepsilon\}$. By independence (Lemma 2.1) and Definition 4.7,

$$\mathbb{E}[e^{tS_n}] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] \leq e^{Vt^2/2}, \quad \text{but only for } |t| \leq 1/\alpha.$$

So the Cramér–Chernoff bound (Theorem 2.1) gives

$$\mathbb{P}(S_n \geq n\varepsilon) \leq \exp\left(-t n\varepsilon + \frac{1}{2} V t^2\right) \quad \text{for } 0 < t \leq 1/\alpha.$$

The exponent is minimized at $t^* = n\varepsilon/V$, but unlike the sub-Gaussian case the constraint $t \leq 1/\alpha$ may rule this out. *This is precisely what creates the two regimes.*

Small deviations ($\varepsilon \leq V/(\alpha n)$). Here $t^* \leq 1/\alpha$ is admissible, and substituting $t = t^*$:

$$-t^* n\varepsilon + \frac{1}{2} V (t^*)^2 = -\frac{n^2 \varepsilon^2}{2V} = -\frac{n\varepsilon^2}{2V/n},$$

which is the first bound. The sub-exponential tail is never felt, and we recover a Gaussian-type bound driven by the variance proxy V/n .

Large deviations ($\varepsilon > V/(\alpha n)$). Now $t^* > 1/\alpha$. The exponent is decreasing in t on $[0, t^*]$, so the best admissible choice is the boundary $t = 1/\alpha$:

$$\mathbb{P}(S_n \geq n\varepsilon) \leq \exp\left(-\frac{n\varepsilon}{\alpha} + \frac{V}{2\alpha^2}\right).$$

The regime assumption gives $V < \alpha n\varepsilon$, hence $V/(2\alpha^2) < n\varepsilon/(2\alpha)$, and the exponent is at most $-n\varepsilon/\alpha + n\varepsilon/(2\alpha) = -n\varepsilon/(2\alpha)$. This is the second bound: once the deviation is large enough, the MGF is only controlled up to $|t| \leq 1/\alpha$, and the tail decays merely *exponentially* in ε rather than in ε^2 . $\square$

Example 4.9 — χ^2 Distribution is Sub-Exponential

Let $Z \sim \mathcal{N}(0, 1)$ and $X = Z^2 - 1$, so $\mathbb{E}[X] = 0$. This is the building block of the χ^2 distribution: if $X_1, \dots, X_n$ are i.i.d. copies of X , then $\sum_{i=1}^n X_i = \chi_n^2 - n$.

Step 1: the MGF. Since $\mathbb{E}[e^{tZ^2}] = (1 - 2t)^{-1/2}$ for $t < 1/2$,

$$\mathbb{E}[e^{tX}] = e^{-t}(1 - 2t)^{-1/2}, \quad \psi(t) := \log \mathbb{E}[e^{tX}] = -t - \frac{1}{2} \log(1 - 2t).$$

Note $\mathbb{E}[e^{tX}] = \infty$ for $t \geq 1/2$: the MGF exists only on a bounded interval, which is exactly why X cannot be sub-Gaussian.

Step 2: bounding ψ near the origin. Expanding $-\frac{1}{2} \log(1 - 2t) = \sum_{k \geq 1} \frac{2^{k-1}}{k} t^k$, the linear terms cancel and

$$\psi(t) = \sum_{k \geq 2} \frac{2^{k-1}}{k} t^k.$$

For $|t| \leq 1/4$, bound $1/k \leq 1/2$ and sum the geometric series:

$$\psi(t) \leq \frac{1}{4} \sum_{k \geq 2} (2|t|)^k = \frac{t^2}{1 - 2|t|} \leq 2t^2,$$

using $1 - 2|t| \geq 1/2$ in the last step. Thus $\mathbb{E}[e^{tX}] \leq e^{4t^2/2}$ for $|t| \leq 1/4$, i.e. $X \sim \text{subE}(4, 4)$ — that is, $\nu^2 = 4$ and $\alpha = 4$.

Step 3: concentration for χ_n^2 . Apply Theorem 4.8 with $\nu_i^2 = 4$ and $\alpha = 4$, so $V = 4n$ and the regime threshold is $V/(\alpha n) = 1$. Since $\frac{1}{n} \sum_i X_i = \chi_n^2/n - 1$, we get

$$\mathbb{P}(\chi_n^2 \geq n(1 + \varepsilon)) \leq \begin{cases} e^{-n\varepsilon^2/8}, & 0 \leq \varepsilon \leq 1 \quad (\text{Gaussian regime}), \\ e^{-n\varepsilon/8}, & \varepsilon > 1 \quad (\text{exponential regime}). \end{cases}$$

The crossover sits exactly at $\varepsilon = 1$: deviations comparable to the mean n itself.

A sharper packaging. The classical Laurent–Massart bound

$$\mathbb{P}(\chi_n^2 \geq n + 2\sqrt{n\varepsilon} + 2\varepsilon) \leq e^{-\varepsilon}$$

puts both regimes into a single statement. Writing the deviation as $u = 2\sqrt{n\varepsilon} + 2\varepsilon$: when $\varepsilon \ll n$ the term $2\sqrt{n\varepsilon}$ dominates, and inverting gives the Gaussian tail $e^{-u^2/(4n)}$; when $\varepsilon \gg n$ the term 2ε dominates, giving the exponential tail $e^{-u/2}$.

5 Tightness, Limitations, and Comparison with CLT

5.1 How tight are these bounds?

Remark 5.1 — Tightness of Hoeffding's Inequality

For $X_i \stackrel{\text{iid}}{\sim}$ Rademacher (± 1 with equal probability), Hoeffding gives $\mathbb{P}(\bar{X}_n \geq \varepsilon) \leq e^{-n\varepsilon^2/2}$. By the Berry–Esseen theorem, the true probability satisfies

$$\mathbb{P}(\bar{X}_n \geq \varepsilon) = \mathbb{P}(\mathcal{N}(0, 1) \geq \varepsilon\sqrt{n}) + O(n^{-1/2}) \approx \frac{1}{\varepsilon\sqrt{2\pi n}} e^{-n\varepsilon^2/2}.$$

The Hoeffding bound is *tight in the exponent* but misses the polynomial prefactor $1/(\varepsilon\sqrt{n})$. For the purposes of guaranteeing confidence (δ chosen a priori), the exponent is what matters.

5.2 The price of distribution-freeness

	CLT	Hoeffding	Bernstein
Assumptions	i.i.d., $\sigma^2 < \infty$	Independence, bounded	Independence, bounded or sub-exp
Validity	Approximate, large n	Exact, all n	Exact, all n
Tail type	Gaussian (from CLT)	Gaussian in exponent	Gaussian/Poisson regime
Uses variance?	Yes (σ^2)	No (range only)	Yes (σ_i^2)
Sample size (Example 3.3)	$n \approx 385$	$n \approx 530$	≈ 385 (near variance)

Remark 5.2 — When Is Each Bound Best?

- Use **CLT** when n is large, the distribution is well-behaved (finite variance, not too heavy-tailed), and you need a tight, interpretable confidence interval.
- Use **Hoeffding** when n is moderate, the variable is bounded, you need a *guaranteed* (non-approximate) confidence bound, or the distribution is completely unknown.
- Use **Bernstein** when $\sigma^2 \ll (b - a)^2$ (rare events, highly skewed distributions) or when the Hoeffding bound is vacuous.
- **Sub-Gaussian framework** unifies Gaussian and Hoeffding bounds; use it when analyzing algorithms (e.g., multi-armed bandits, randomized algorithms) where inputs may have various distributions.

5.3 Limitations of non-asymptotic bounds

Remark 5.3 — What Non-Asymptotic Bounds Cannot Do

Non-asymptotic concentration inequalities have their own limitations:

- **Looseness:** The bounds can be far from tight in constant factors, especially when the distribution is Gaussian and n is large. In that regime, the CLT-based interval is sharper.
- **Bounded support required:** Hoeffding requires bounded variables. For unbounded but sub-Gaussian variables, the framework extends, but the parameter σ must be known or estimated.
- **Dependence:** All bounds here require independence. Extensions to martingales (Azuma–Hoeffding) and mixing sequences exist but require more structure.

6 Summary

1. The **Cramér–Chernoff method**: apply Markov to e^{tX} , bound the MGF, optimize over t . This three-step recipe produces all bounds in this lecture.
2. **Hoeffding’s lemma** bounds $\log \mathbb{E}[e^{tX}] \leq t^2(b-a)^2/8$ for bounded centered variables via convexity.
3. **Hoeffding’s inequality**: for independent bounded variables, $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2/(b-a)^2)$. Valid for *every* n , no distributional assumption beyond boundedness.
4. **Bernstein’s inequality** incorporates the variance σ^2 , giving a tighter bound (particularly for rare-event or heavy-tailed regimes) via a two-regime structure.
5. The **sub-Gaussian family** $\text{subG}(\sigma^2)$ unifies Gaussians and bounded variables under one framework with four equivalent characterizations. Sub-Gaussianity is closed under independent sums.
6. All bounds are *exact* (non-approximate), *distribution-free* (beyond the stated assumptions), and *dimension-free* for scalar problems. They are tight in the exponent but may miss polynomial prefactors.

Looking ahead. Lecture 4 applies these tools to simultaneous inference via the union bound (the $\sqrt{\log p}$ price of multiplicity) and to the maximum of sub-Gaussian variables, revisits the bootstrap as a middle ground between the two paradigms, and develops a unified framework for deciding which paradigm to use in practice.

7 Exercises

Exercise 7.1 (Cramér–Chernoff for Poisson). *Let $X \sim \text{Poisson}(\lambda)$, so $M_X(t) = \exp(\lambda(e^t - 1))$.*

- (a) *Apply the Cramér–Chernoff bound to $\mathbb{P}(X \geq k)$ for $k > \lambda$. Optimize over $t > 0$ and show the bound is $\left(\frac{e\lambda}{k}\right)^k e^{-\lambda}$.*
- (b) *Compare this bound with Markov’s inequality $\mathbb{P}(X \geq k) \leq \lambda/k$. For which values of k is the Cramér–Chernoff bound tighter?*

Exercise 7.2 (Hoeffding vs. Chebyshev). *Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1]$. Fix $\varepsilon = 0.05$ and $\delta = 0.01$.*

- (a) *Find the minimum n such that $\mathbb{P}(|\bar{X}_n - \frac{1}{2}| \geq \varepsilon) \leq \delta$ using Chebyshev’s inequality.*
- (b) *Find the minimum n using Hoeffding’s inequality.*
- (c) *Find the minimum n using the CLT approximation.*
- (d) *Rank the three bounds from most to least conservative and explain why.*

Exercise 7.3 (Hoeffding’s Lemma: the constant $1/8$). (a) *In the proof of Hoeffding’s Lemma, verify that $\varphi''(u) = \gamma(1 - \gamma)e^u / (1 - \gamma + \gamma e^u)^2 \leq 1/4$ for all $u \in \mathbb{R}$ and $\gamma \in [0, 1]$. (Hint: use AM-GM on $\gamma e^{u/2}$ and $(1 - \gamma)e^{-u/2}$.)*

- (b) *Show the constant $1/8$ is achieved in the limit $\gamma \rightarrow 1/2$.*

Exercise 7.4 (Sub-Gaussian characterizations). *Let $X \sim \text{subG}(\sigma^2)$.*

- (a) *Show that $\mathbb{E}[X^{2k}] \leq (2\sigma^2)^k k!$ for all integers $k \geq 1$. (Hint: expand $e^{t^2\sigma^2/2}$ in a power series and compare with the power series of $\mathbb{E}[e^{tX}]$.)*
- (b) *Deduce that $(\mathbb{E}[|X|^k])^{1/k} \leq C\sigma\sqrt{k}$ for a universal constant C .*
- (c) *Give an example showing that a $\text{subG}(\sigma^2)$ variable need not be bounded.*

Exercise 7.5 (Bernstein’s advantage). *Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $p = 0.001$ and $n = 50,000$.*

- (a) *Show that the Hoeffding bound on $\mathbb{P}(\bar{X}_n \geq 2p)$ exceeds 1.*
- (b) *Compute the Bernstein bound on $\mathbb{P}(\bar{X}_n \geq 2p)$ (use $M = 1$, $\sigma^2 = np(1 - p)$, and replace ε by np).*
- (c) *Compare with the exact Poisson approximation: $S_n \approx \text{Poisson}(np)$, so $\mathbb{P}(S_n \geq 2np) \approx e^{-np}(1 + np + \dots)^{-1}$.*

Exercise 7.6 (McDiarmid’s inequality (extension)). *Let $f(X_1, \dots, X_n)$ be a function of independent random variables satisfying the bounded differences condition:*

$$\sup_{x_1, \dots, x_n, x'_i} |f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

The **McDiarmid inequality** states $\mathbb{P}(f - \mathbb{E}[f] \geq \varepsilon) \leq \exp(-2\varepsilon^2 / \sum_i c_i^2)$.

- (a) Verify that when $f(X_1, \dots, X_n) = \bar{X}_n$ and $X_i \in [a, b]$, McDiarmid reduces to Hoeffding.
- (b) Apply McDiarmid to bound the fluctuation of the empirical Rademacher complexity $\hat{\mathcal{R}}_n(\mathcal{F})$ when each X_i is changed by one sample. (This result is used in statistical learning theory in Lecture 4.)

References

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